

ASSIGNMENT No: 3

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TITLE: Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

Problem Statement: Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

Learning Objectives: To implement method overloading and use static variables and methods in Java programs.

Theory: Method overloading allows multiple methods to have the same name with different parameter lists, enabling compile-time polymorphism. The static keyword is used for variables and methods that belong to the class instead of individual objects. Static members share common data among all objects and reduce memory usage. These concepts improve code flexibility, readability, and efficient resource management.

Exercise

1. Develop a Calculator program using overloaded methods for addition of integers and decimals. Use a static variable to count calculations.

Code & Output:

```
J Calculator.java > Calculator
1 //Assignment 3
2 public class Calculator{
3     int add(int a, int b){
4         return a + b;
5     }
6     float add(float a, float b){
7         return (a + b);
8     }
9     int add(int a, int b, int c){
10        return (a + b + c);
11    }
12    Run | Debug
13    public static void main(String[] args){
14        Calculator calc = new Calculator();
15        System.out.println(calc.add(a: 5, b: 10));
16        System.out.println(calc.add(a: 5.5f, b: 10.5f));
17        System.out.println(calc.add(a: 5, b: 10, c: 15));
18    }
19 }
```

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
[Running] cd "c:\Users\arhaa\OneDrive\Desktop\java\" && javac assignment2.java && java assignment2
Name: Bob
Age: 95

[Done] exited with code=0 in 0.922 seconds

[Running] cd "c:\Users\arhaa\OneDrive\Desktop\java\" && javac Calculator.java && java Calculator
15
16.0
30
```

2. Develop a Restaurant Billing Application where overloaded methods calculate bills for dine-in, takeaway, and delivery orders, while static variables track total orders.

Code & Output:

J Restaurant.java > Restaurant > main(String[])

```
1 public class Restaurant {
2     static int totalOrders = 0;
3
4     // Dine-in bill
5     void bill(double amount) {
6         totalOrders++;
7         System.out.println("Dine-In Bill: Rs." + amount);
8     }
9
10    // Takeaway bill
11    void bill(double amount, double packingCharge) {
12        totalOrders++;
13        System.out.println("Takeaway Bill: Rs." + (amount + packingCharge));
14    }
15
16    // Delivery bill
17    void bill(double amount, double packingCharge, double deliveryCharge) {
18        totalOrders++;
19        System.out.println("Delivery Bill: Rs." +
20            (amount + packingCharge + deliveryCharge));
21    }
22
23    Run | Debug
24    public static void main(String[] args) {
25        Restaurant r = new Restaurant();
26
27        r.bill(amount: 500); // Dine-in
28        r.bill(amount: 500, packingCharge: 20); // Takeaway
29        r.bill(amount: 500, packingCharge: 20, deliveryCharge: 50); // Delivery
30
31        System.out.println("Total Orders: " + totalOrders);
32    }
33
```

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\arhaa\OneDrive\Desktop\java> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enab
ng\Code\User\workspaceStorage\02e7f2cfce92a0ba8e30606ade6b2c24\redhat.java\jdt_ws\java_8f7d62e9\bin
Dine-In Bill: Rs.500.0
Takeaway Bill: Rs.520.0
Delivery Bill: Rs.570.0
Total Orders: 3
PS C:\Users\arhaa\OneDrive\Desktop\java>
```